u_values

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1 U-Values and Thermal Calculations

This notebook calculates thermal conductivity (lambda) values, R-values, and U-values for building materials, with a focus on External Wall Insulation (EWI) payback analysis.

1.1 Setup: Import Libraries

[2]:
```
%pip install pint
import pint

ureg = pint.UnitRegistry()
# ureg.define('gbp = []')  # Define GBP as a dimensionless currency unit
ureg.define('gbp = [currency]')
ureg.gbp.symbol = '£'
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pint in /home/steve/.local/lib/python3.14/site-
packages (0.25.3)
Requirement already satisfied: flexcache>=0.3 in
/home/steve/.local/lib/python3.14/site-packages (from pint) (0.3)
Requirement already satisfied: flexparser>=0.4 in
/home/steve/.local/lib/python3.14/site-packages (from pint) (0.4)
Requirement already satisfied: platformdirs>=2.1.0 in /usr/lib/python3.14/site-
packages (from pint) (4.2.2)
Requirement already satisfied: typing-extensions>=4.0.0 in
/usr/lib/python3.14/site-packages (from pint) (4.15.0)
Note: you may need to restart the kernel to use updated packages.
```

1.2 Material Properties: Thermal Conductivity (Lambda)

Units: W/mK (watts per metre-kelvin)

[3]:
```
LAMBDA = {
    "xps": 0.04,  # Polyisocyanurate or extruded polystyrene
    "brick": 0.77,  # Medium density clay brick (~1800 kg/m³)
    "clay_brick_lightweight": 0.5,  # Lightweight clay brick (< 1700 kg/m³)
    "plasterboard": 0.25,  # Standard plasterboard (gypsum)
    "plaster": 0.77 # assumed
    # add a few more here when everything else works
```

```
}
```

1.3 Wall Construction Definition

Thicknesses in meters

1.4 R and U from thickness and conductivity

[4]:
```
def lambda_val(material):
    """Get thermal conductivity with units."""
    return LAMBDA[material] * ureg.W / (ureg.m * ureg.kelvin)

def R_value(element, thickness):
    """Calculate R-value (thermal resistance) for a wall element."""
    return  thickness/ lambda_val(element)

def U_value(wall):
    """
    The U value of the wall, in SI units.
    """
    units_of_U = ureg.watt / ureg.meter ** 2 / ureg.K
    tot_R = 0.0 / units_of_U
    for i in wall:
        R = R_value(i,wall[i] * ureg.meters)
        tot_R += R
    return 1./tot_R
```

[5]:
```
assert lambda_val('plaster') == 0.77 * ureg.W / ureg.m / ureg.K
```

[6]:
```
lambda_val('xps')
```

[6]: $0.04 \frac{\text{watt}}{(\text{kelvin}\cdot\text{meter})}$

1.5 Calculate Total R-Value and U-Value for Wall

[7]:
```
u_with_EWI = U_value(
    {
    "plasterboard": 0.013,  # Internal plaster finish
    "brick": 0.220,  # Main structural brick layer
    "xps": 0.0700,})  # Internal insulation board

print(f"u_with_EWI: {u_with_EWI:.2fP~}")
```

```
u_with_EWI: 0.48 W/K/m²
```

[28]:
```
u_without_EWI = U_value(
    {
```

```python
    "plaster": 0.025,  # Internal plaster finish
    "brick": 0.2200,  # Main structural brick layer
    })

print(f"u_without_EWI: {u_without_EWI:.2fP~}")
```

```
u_without_EWI: 3.14 W/K/m²
```

1.6 External Wall Insulation (EWI) Analysis

1.6.1 Calculate R-values for EWI Components

1.6.2 Energy and Cost Calculations

[37]:
```python
# Constants

degree_days = 2250 * ureg.K * 60*60*24 * ureg.s  # Assumed for Knebworth (base 15°C)
print(f"degree days (in SI units!) {degree_days:.2eP~}")
```

```
degree days (in SI units!) 1.94×10  K·s
```

[38]:
```python
# gas price, as quoted on your gas bill:
gas_price = 0.03*ureg.gbp / ((1000 * ureg.W) * (60*60*ureg.s))   # gbp/kWh (2021 pre-crisis price)
print(f"gas price {gas_price:.2eP~}")
```

```
gas price 8.33×10   gbp/s/W
```

[39]:
```python
# Work out annual cost

# base temperate should be about 15C.
# degree days should be about 2500-600 (if base temp is 15C).

# Calculate U-value improvement
delta_u = u_without_EWI - u_with_EWI

print(f"Change in U-value from EWI: {delta_u:.4fP~}")
```

```
Change in U-value from EWI: 2.6639 W/K/m²
```

[41]:
```python
# Calculate unit savings
saving = gas_price * delta_u # per unit time
print(f"saving: {saving:.6eP~}")
saving_pd = saving*60*60*24*ureg.s
print(f"saving pd: {saving_pd:.2eP~}")
# print(f"saving on a 15K temp delta: {saving_pd*15*ureg.K}")

year_cost_psm = degree_days  * gas_price * delta_u
```

```python
print(f"Annual cost saving: {year_cost_psm:.2fP~}")
```

```
saving: 2.219887×10  gbp/K/m²/s
saving pd: 1.92×10 ³ gbp/K/m²
Annual cost saving: 4.32 gbp/m²
```

1.6.3 Payback Calculation

[43]:
```python
# Installation cost
# Based on non-binding verbal quote from TAW: gbp30K for front and side␣
↪elevations (~150m²)
installation_cost = 150 * ureg.gbp / (ureg.meter**2)  # gbp/m²

print(f"Installation cost: {installation_cost:.0fP~}")
print(f"Annual saving per m²: {year_cost_psm:.2fP~}")

simple_payback = installation_cost / year_cost_psm
print(f"Simple payback period: {simple_payback:.1f} years")
```

```
Installation cost: 150 gbp/m²
Annual saving per m²: 4.32 gbp/m²
Simple payback period: 34.8 dimensionless years
```

1.7 Summary

- **Original wall U-value**: f"~{:.2f}"
- **Wall with EWI U-value**: ~{:.2f} W/m²K
- **Estimated payback**: ~{:.0f} years (at 2021 gas prices)

Note: Payback period will be shorter with current higher gas prices.